TIMO GÜNTHER

Data Science | AI | IT Projects

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Based in Würzburg, Germany | Relocating to Cologne

PROFILE

Master's degree in Digital Humanities with a machine learning thesis (grade 1.0, the top grade) on XGBoost and transformer-based NLP, plus NLP projects with BERT (Hugging Face) and Word2Vec. Around two and a half years of experience as a consultant in IT client projects (implementing SAP IBP for mostly mid-sized companies), from client workshops through data model and data integration to testing and training. Hands-on skills in Python, SQL, Pandas, Scikit-learn and PyTorch. I am looking for a role in data science, AI and digital projects that brings together my studies and my consulting experience.

WORK EXPERIENCE

Consultant
Feb 2024 - Sep 2026

TIVTAL GmbH, Würzburg | specialised IT consultancy for planning software (SAP IBP)

- Captured and analysed clients' planning processes, ran client workshops and translated requirements into functional and solution concepts
- Configured and implemented the software, including the data model (planning areas, master data types, key figures), planning views, dashboards and alerts
- Set up and evaluated statistical forecasting methods as part of the team; reported forecast accuracy in key figures (e.g. MAPE)
- Connected source systems (SAP S/4HANA, ECC) via SAP Cloud Integration; modelled integration flows and mapped master and transactional data
- Wrote test cases, planned and supported integration and user acceptance tests, including error analysis and fixes
- Delivered key user and end user training including training materials, supported go-lives and hypercare phases, and provided ongoing support (application management, change requests, release upgrades)
- Used AI to automate recurring tasks: a Python program, developed with ChatGPT, that compares authorisations between Excel and SAP, and a Claude skill shared with the whole team that produces presentation-ready PowerPoint slides

Administrative Employee (part-time, alongside studies)
Dec 2018 - Jan 2024

Service Centre InterNational Transfer (International Office), University of Würzburg | 15-20 hrs/week

- Maintained the internationalisation databases; analysed and visualised data on mobility and international exchange programmes
- Maintained and updated the websites with the TYPO3 CMS, researched partner universities online, handled enquiries and mobility applications
- Co-organised large international events, including the JMU International Evening (approx. 3,000 attendees) and the annual conference of the CHARM-EU university alliance (approx. 250 attendees)

Technology Advisor (working student)
Oct 2017 - Nov 2018

DTB Deutsche Technikberatung GmbH | service start-up for home technology support (CECONOMY group)

- Advised customers in person, delivered individual on-site training and technical support (installation, troubleshooting, set-up)
- Independently analysed customers' technology set-ups and gave practical recommendations; wrote documentation and final reports

EDUCATION

M.A. Digital Humanities Apr 2020 - Oct 2023

Julius-Maximilians-Universität Würzburg | final grade 1.6 (German scale, 1.0 is the top grade)

- Master's thesis (grade 1.0): “Comparative Analysis of Data Sets and Feature Importance in Predicting Hit Songs”, a machine learning analysis for predicting hit songs with XGBoost and transformer-based NLP (sentiment and emotion analysis of the lyrics) in Python; code and thesis: github.com/tig1795/predicting_hits_ML_MA
- Focus areas included data modelling and databases (SQL); electives: Human-Computer Interaction (1.0), Usability and Software Ergonomics (1.3)

B.A. Digital Humanities and History Oct 2015 - Mar 2020

Julius-Maximilians-Universität Würzburg | final grade 2.2

- Bachelor's thesis (grade 1.7, written in German): “Elections in the Weimar Republic 1919 to 1933. The influence of economic development on voting behaviour”, with the election results mapped in QGIS
- Focus areas included XML data processing

Economics (no degree) Oct 2014 - Jul 2015

Julius-Maximilians-Universität Würzburg

Abitur (university entrance qualification), STEM focus 2014

Hanns-Seidel-Gymnasium, Hösbach

PROJECTS (UNIVERSITY)

Semantic change of words with BERT and Word2Vec

University of Würzburg | Python, Hugging Face (BERT), Word2Vec | github.com/tig1795/word_embeddings_Lexika

- Investigated whether BERT is suitable for detecting semantic change, using pre-trained Hugging Face models (including bert-base-german-cased and a historical German BERT) compared with Word2Vec embeddings; fine-tuned the historical German BERT model with the Hugging Face Trainer

Clustering and classification of 2,225 BBC news articles

University of Würzburg | Python, Kaggle dataset | github.com/tig1795/BBC_News_cluster

- Preprocessing (stemming, POS tagging), K-means, hierarchical clustering, GMM, DBSCAN, SVM classification, topic modelling (LDA, NMF), Word2Vec and an LSTM autoencoder with TensorFlow/Keras; evaluated with the silhouette coefficient and AIC/BIC, among others

Usability comparison of two degree programme websites (group project)

University of Würzburg | heuristic evaluation, cognitive walkthrough, user tests (AttrakDiff, UEQ)

- Compared the websites of two degree programmes from the students' perspective; took on a large part of the project organisation, including regular coordination with the supervisor

CERTIFICATES & TRAINING

SAP Certified - SAP Integrated Business Planning | SAP | obtained 05/2025, valid until 04/2027

TYPO3 Basics | course with practical exercises, University of Würzburg computing centre | 03/2021

SKILLS

Data science & programming: Python, SQL, Pandas, Scikit-learn, PyTorch, TensorFlow, XGBoost, SPSS

NLP & AI: Hugging Face (BERT, transformer models), Word2Vec, ChatGPT, Claude (incl. skill development)

Methods & tools: client workshops, requirements analysis, test cases, user training, Jira, GitHub, Kaggle

Integration & systems: SAP Cloud Integration (CI-DS), SAP S/4HANA and ECC, SAP IBP (certified)

Data & web: databases, XML, EPUB, QGIS, TYPO3 (CMS), HTML/CSS, PHP, JavaScript (basics)

Languages: German (native), English (B2+/C1), French (B1+)

VOLUNTEERING & INTERESTS

Tutor at Hanns-Seidel-Gymnasium | volunteer at the Kommz festival | Sportfreunde Sailauf (football club)

Football, gaming, motorsport, music